

RAJ TEJPAL KHATIK

AI/ML Engineer | Agentic AI Systems | GenAI Research

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PROFILE

AI/ML Engineer and MSc Applied AI candidate at the University of Warwick, with two years' prior industry experience in data analytics and ML engineering. Designs multi-agent, retrieval-augmented systems that reason through causal chains — six deployed projects, four preprints live on SSRN. Seeking AI/ML Engineer, GenAI Engineer, and Agentic AI roles.

KEY SKILLS

Python, FastAPI, LangGraph, LangChain, JavaScript, React · RAG, FAISS, Agentic AI, GenAI, Prompt Engineering, LLM Evaluation, Statistical Testing · NumPy, Pandas, SQL/SQLite, Power BI · Streamlit Cloud, Cloudflare Pages/Workers, Git/GitHub

EXPERIENCE

Play Tester — Fusion UX Lab, University of Warwick

Nov 2025 – Aug 2026

- Evaluate video games and interactive systems through structured playtesting; provide user-centred feedback on usability, player experience, and mechanics.

AI Engineer → Software Engineer → Software Engineer Intern — H&B Infotech, India

Aug 2022 – Mar 2025

- Built ML pipelines for model training data prep, integrated into backend systems — improving analysis accuracy and reporting efficiency.
- Engineered backend systems, data pipelines and Python-MySQL modules for internal reporting tools, with front-end integration.

SELECTED PROJECTS

- SemiBot (Flagship) — 18-factor causal-chain RAG for semiconductor equities; accuracy scales 41.7%→84.0% with market cap ($p < 0.001$, $n = 426$). LangGraph, FAISS, GPT-4o-mini. Presented at Warwick.
- GoldBot — 9-factor gold safe-haven classifier stress-tested across 4 regimes/18 events; diagnosed trend-matching vs. genuine event discrimination.
- SilverBot — Dual-regime industrial/monetary decomposition for silver; industrial signal wins 3/4 regimes ($p = 0.041$), reverses in 2020 crisis.
- Hallucination Detector (Flagship) — 6-signal LLM ensemble (NLI, retrieval grounding, LLM-judge panel); 72.4–92.2% accuracy across 3 benchmarks. Presented at Warwick.
- Retail Research Assistant — Research/comparison platform for all 503 S&P 500 stocks; passed 100/100 adversarial safety test.
- HITL Email Agent — Human-in-the-loop LangGraph email drafter with Gmail OAuth and a Chrome extension (MV3).

EDUCATION

MSc Artificial Intelligence — University of Warwick, WMG

Sept 2025 – Sept 2026

BEng, Computer Science & Engineering — Gujarat Technological University — CGPA 8.49/10

Aug 2021 – May 2025

CERTIFICATIONS & PUBLICATIONS

Certifications: Financial Markets (Yale, Coursera) · GenAI Enhanced Financial Analysis Specialization (Microsoft, Coursera) · Generative AI Engineering (IBM, Coursera)

Publications: 4 preprints live on SSRN (under review) — causal-chain equity analysis, safe-haven asset analysis (gold, silver), and LLM hallucination detection. Full list: portfolio-raj.pages.dev/#papers